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Planning for electric-vehicle evacuations: energy, infrastructure, and storage needs

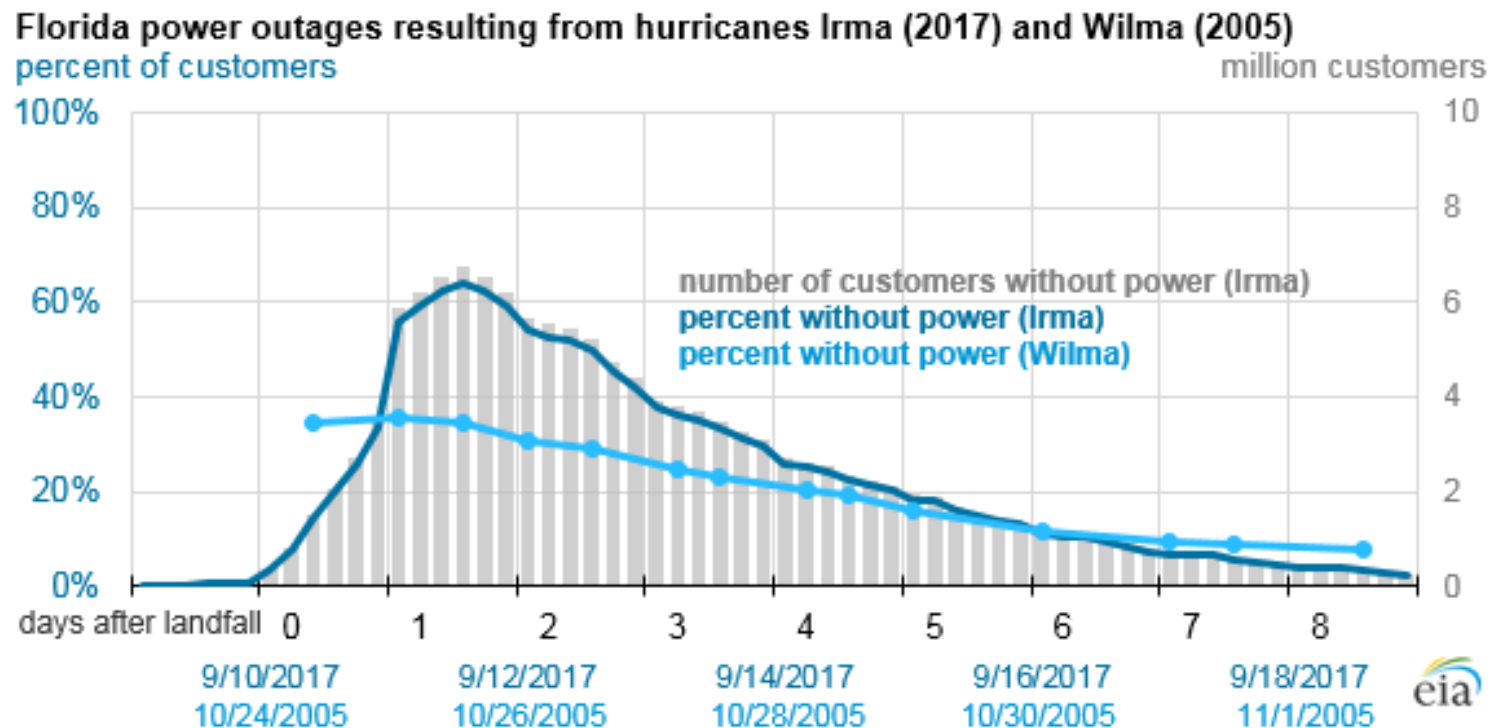
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Motivation: Powering EVs for evacuations

Example of two hurricanes in Florida:

- Power outage due to Hurricane Irma affected nearly two-thirds of Florida's electricity customers¹.
- 90% of people used personal vehicles for evacuations due to Hurricane Irma².



Higher EV adoption requires reliable power solutions, especially during grid failures.

¹ <https://www.eia.gov/todayinenergy/>

² Wong et. al., 2018. Understanding evacuee behavior: a case study of hurricane Irma.



Overview of EV evacuation study

- **Motivation:** Planning for powering EVs for evacuations
- **Part I: Energy needs**
 - **Strategy:** Estimate energy needs for evacuation scenarios using a pragmatic stochastic method.
 - **Results:** Small evacuations (1,000 EVs) over an average distance of 100 km need MWh-level energy; large evacuations (1M EVs) need GWh-level over the same distance.
- **Part II: Charging infrastructure needs**
 - **Strategy:** Assess power requirements of charging infrastructure, comparing with existing infrastructure in some California cities.
 - **Results:** Some California counties have sufficient chargers, but more chargers (especially fast chargers) are crucial for larger and faster evacuations.
- **Part III: Backup storage needs**
 - **Strategy:** Determine backup lithium battery storage size for EV evacuations during grid failure.
 - **Results:** Truck-size backup lithium batteries for small evacuations; football field-size for large ones.

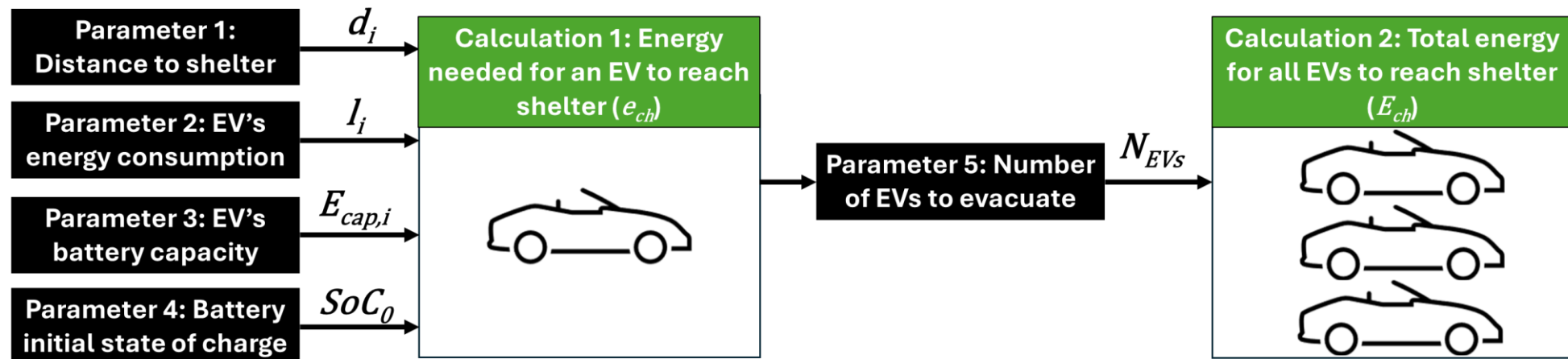


Part I: Energy needs

How much energy does an EV need to evacuate from a danger zone to a safe area?

Part I: Method to estimate energy needs

We treat the parameters as random variables with probabilistic distributions to account for uncertainties.



Mathematical Overview (detailed in the paper):

Energy required to evacuate per vehicle: $e_i = l_i d_i$

Charging energy per vehicle: $e_{ch,i} = \max(e_i - SoC_0 \times E_{cap,i}, 0)$

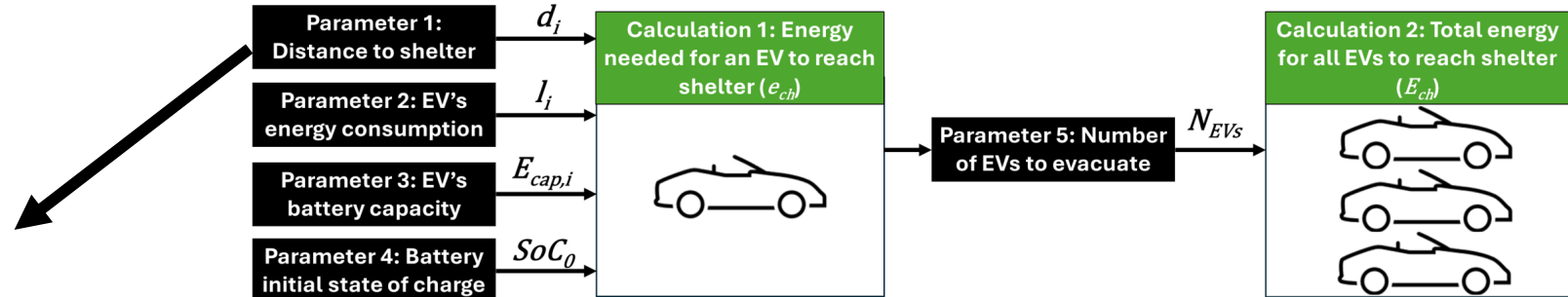
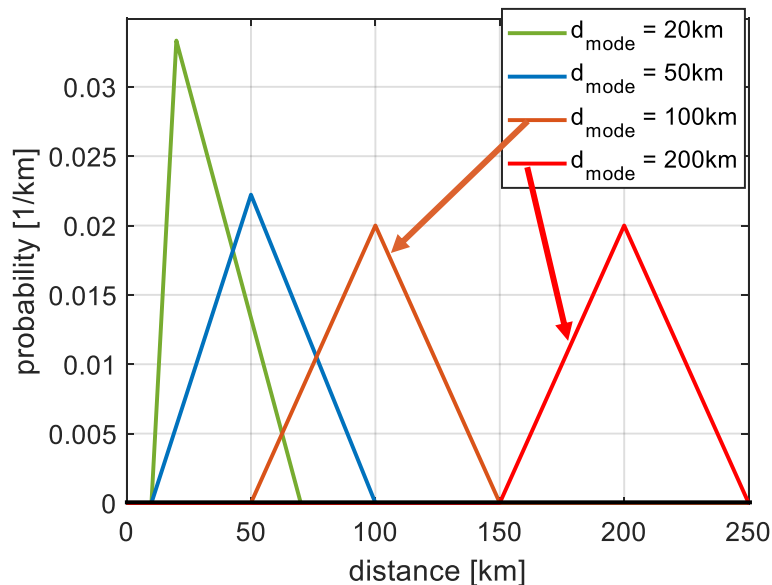
Total charging energy for all EVs: $E_{ch} = \sum_{i=1}^{N_{EV}} e_{ch,i}$

Part I: Method to estimate energy needs

We treat the parameters as random variables with probabilistic distributions to account for uncertainties.

Parameter 1: Distance to shelter, modeled using a triangular distribution $T_d(d_{min}, d_{mode}, d_{max})$

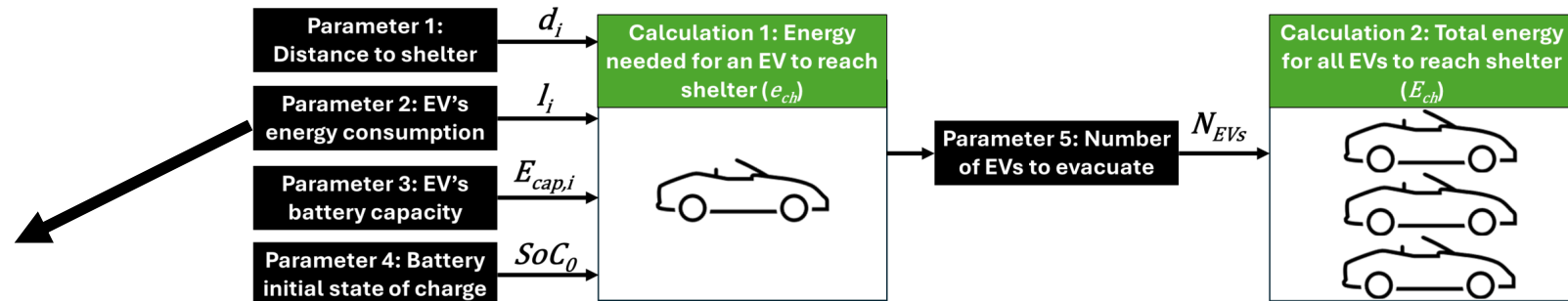
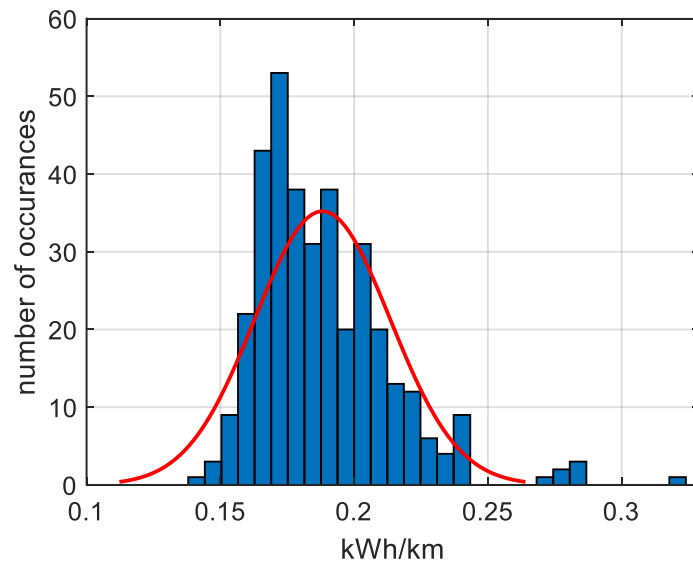
Example:



Part I: Method to estimate energy needs

We treat the parameters as random variables with probabilistic distributions to account for uncertainties.

Parameter 2: Energy consumption (l_i), fitted to a Gaussian distribution based on a histogram of 360 EVs currently on the market¹:



Mean energy consumption: $\mu_l = 0.188$ kWh/km; standard deviation: $\sigma_l = 0.025$ kWh/km

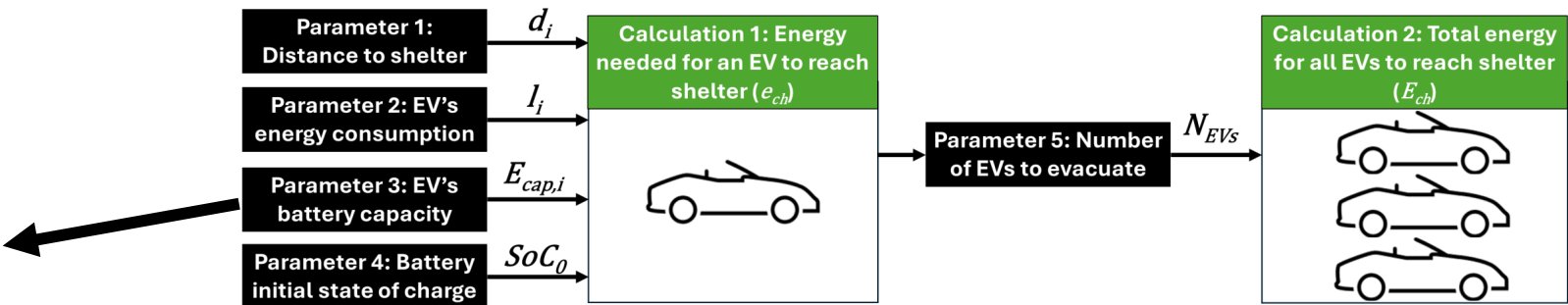
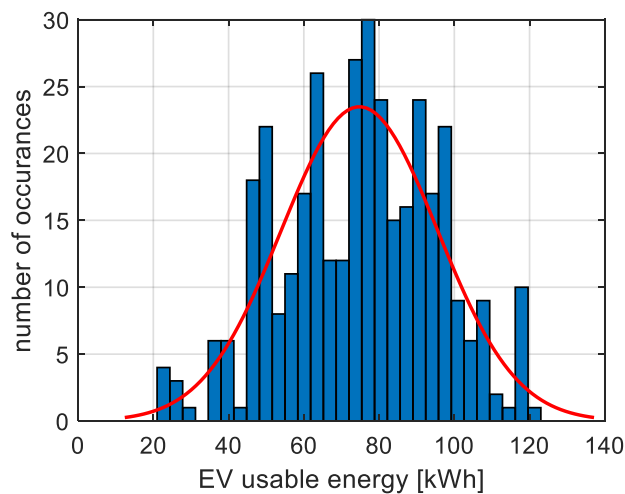
¹ <https://ev-database.org>



Part I: Method to estimate energy needs

We treat the parameters as random variables with probabilistic distributions to account for uncertainties.

Parameter 3: Energy capacity ($E_{cap,i}$), also fitted to a Gaussian distribution from the same 360 EV dataset.



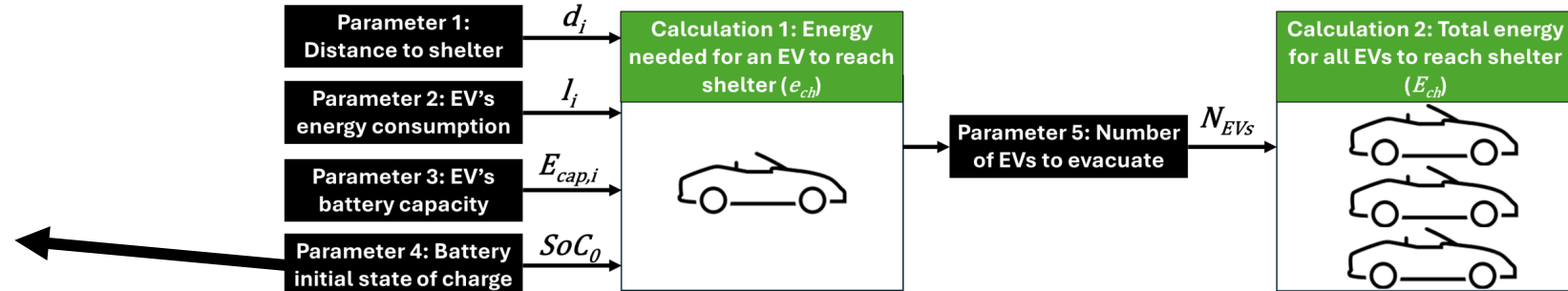
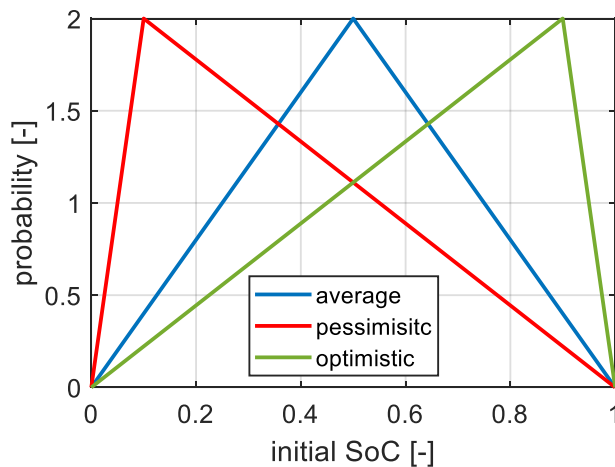
Mean energy capacity: $\mu_l=74.7$ kWh; standard deviation: $\sigma_l= 20.8$ kWh

Part I: Method to estimate energy needs

We treat the parameters as random variables with probabilistic distributions to account for uncertainties.

Parameter 4: Initial state of charge (SoC_0), modeled using a triangular distribution

$SoC_0 \sim \mathbf{T}_{SoC_0}(SoC_{0,min}, SoC_{0,mode}, SoC_{0,max})$:

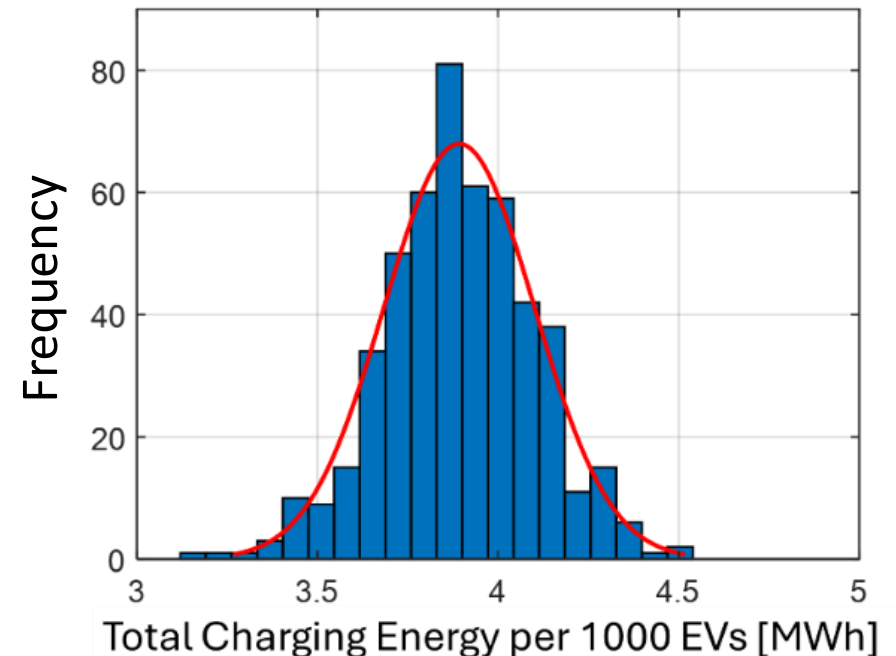


Pessimistic scenario: $SoC_{0,mode} = 0.1$; Average scenario: $SoC_{0,mode} = 0.5$; Optimistic scenario: $SoC_{0,mode} = 0.9$

Part I: Energy needs results

We use a Monte Carlo simulation, sampling random variables from the probabilistic distributions, to estimate the total charging energy:

Example: the result for $d_{mode} = 150$ km, an average initial SoC ($SoC_{0,mode} = 0.5$) and $N_{EV} = 1000$:

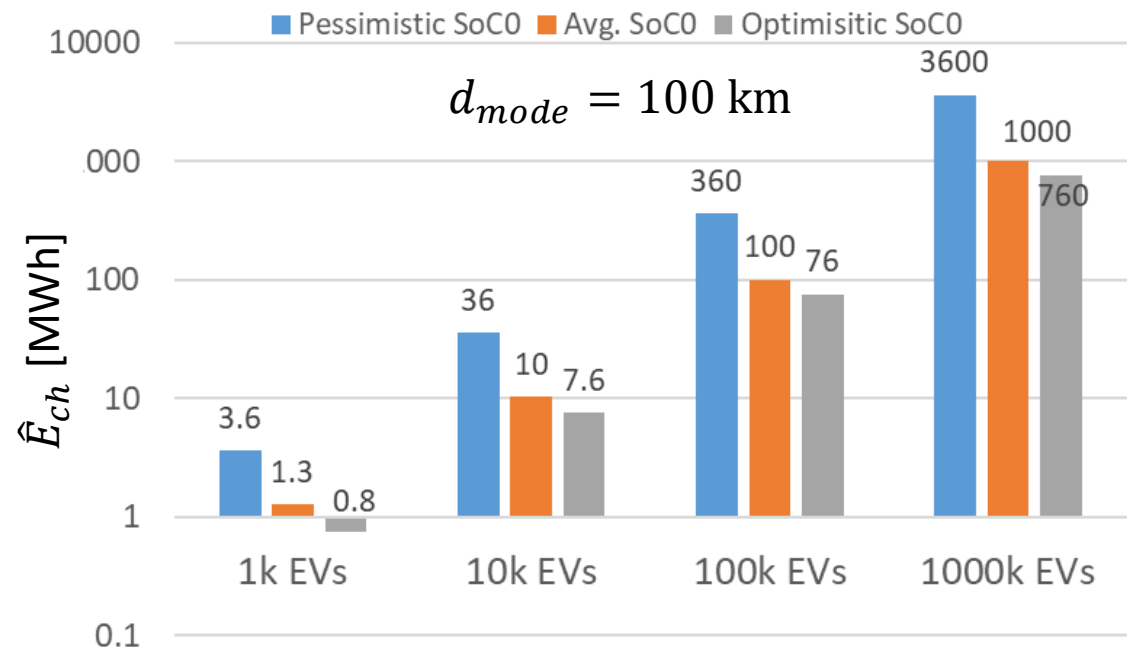
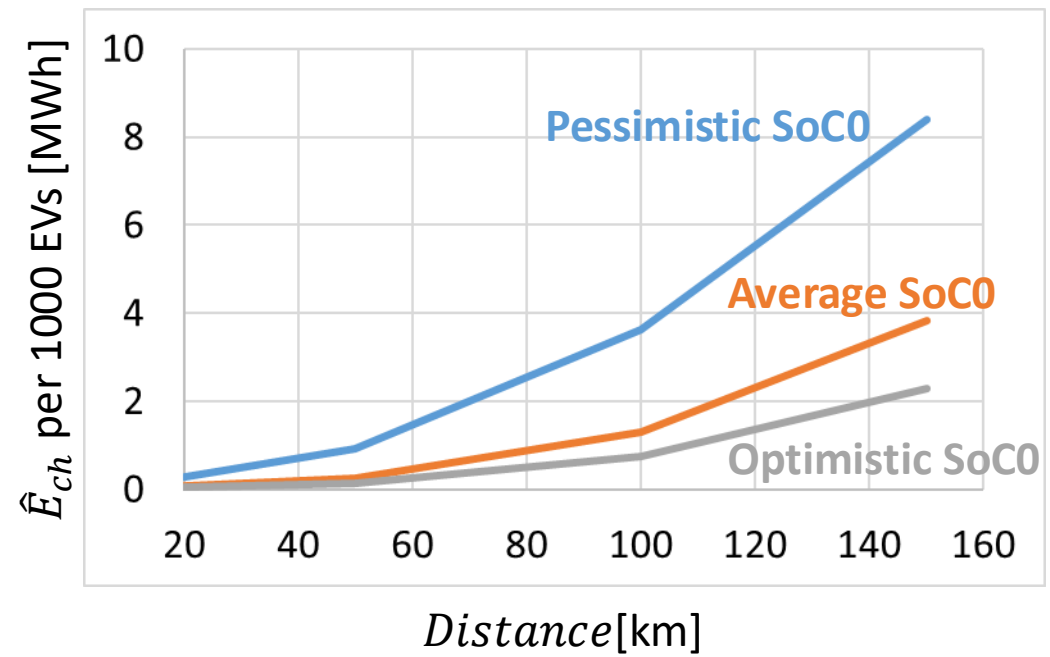


The histogram of E_{ch} closely follows a Gaussian curve. We denote the mean of this Gaussian approximation as $\hat{E}_{ch}$.



Part I: Energy needs results

$\hat{E}_{ch}$ for different evacuation distances, SoC₀, and N_{EVs}:



Evacuating a small neighborhood with 1,000 EVs over an average distance of 100 km requires MWh-level energy, whereas a larger city with 1 million cars typically needs GWh-level energy.



Part II: Charging infrastructure needs

How many Level 2 and fast chargers are needed to meet the power requirements for a specific evacuation scenario?

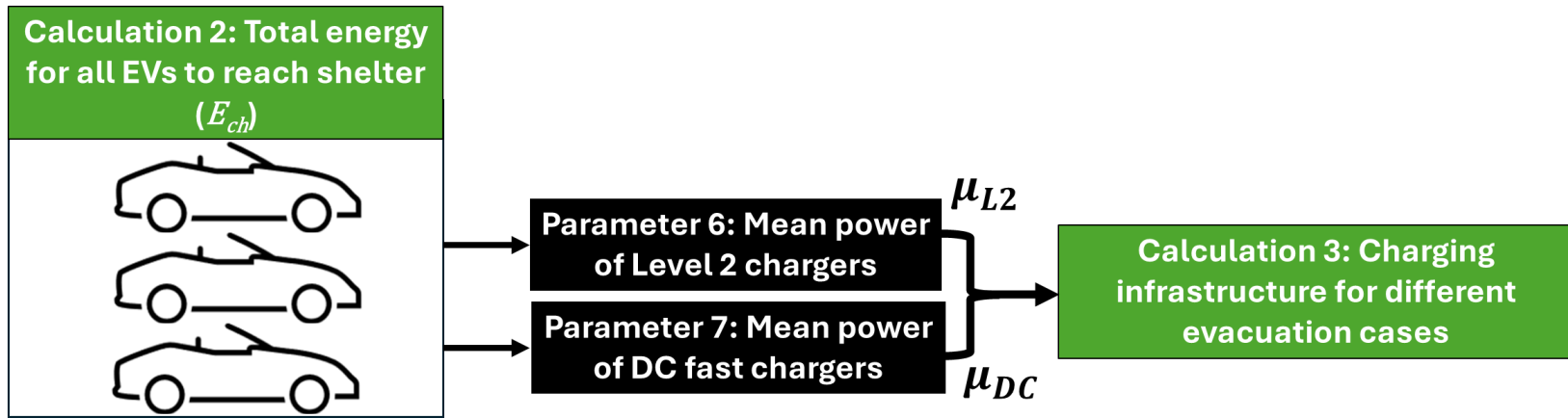


Part II: Method to estimate charging infrastructure needs

The power capability of each individual charger is assumed to have a Gaussian distribution:

Mean power of level-2 chargers: $\mu_{l2}=6$ kW; standard deviation: $\sigma_{l2}= 3$ kW

Mean power of DC fast chargers: $\mu_{DC}=80$ kW; standard deviation: $\sigma_{DC}= 20$ kW



Mathematical Overview (detailed in the paper):

The average charging power needed $\leftarrow p_{CH}^* = \frac{E_{ch}^*}{T_{eva}} \rightarrow$ Charging energy needed
Evacuation time

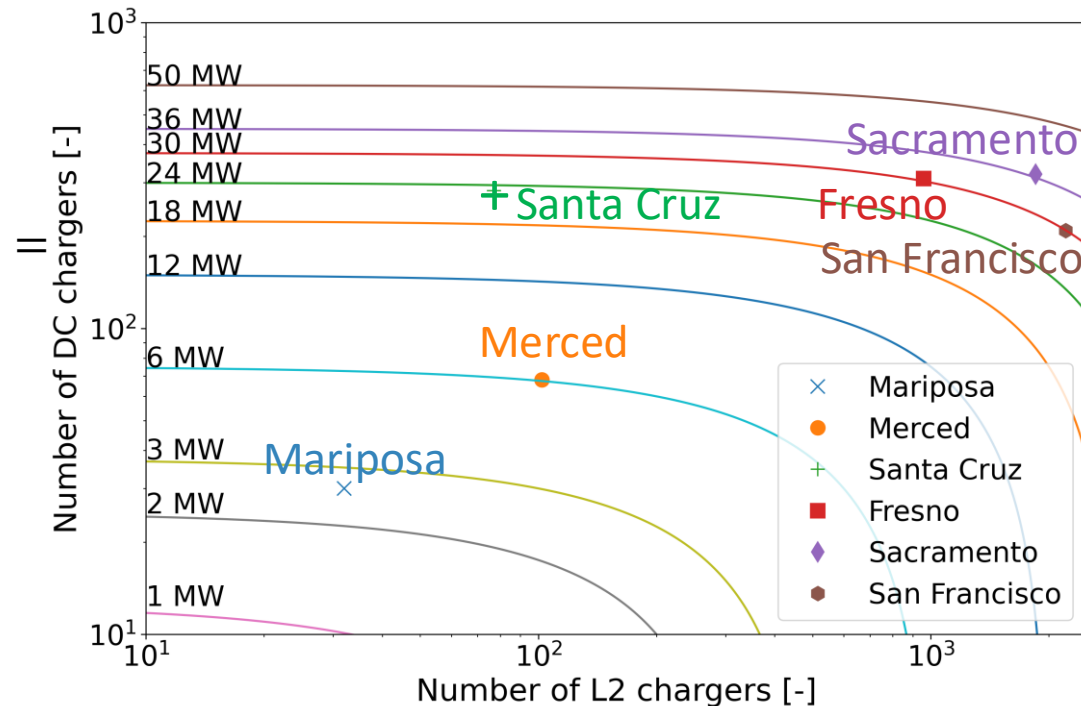
Set of all feasible combinations of Level 2 and DC fast charging stations that can fulfill p_{CH}^* with >50% probability $\leftarrow \hat{\Gamma}(p_{CH}^*) = \{(n_{L2}, n_{DC}): n_{DC}\mu_{DC} + n_{L2}\mu_{L2} \geq p_{CH}^*\}$



Part II: Charging infrastructure needs results

Boundary of $\hat{\Gamma}(p_{CH}^*)$ for different charging power requirements p_{CH}^* :

According to the part I results,
for an average SoC_0 and $d_{mode} = 100$ km and 1k EVs, 1 MWh of charging energy is needed



Assuming an average SoC_0 ,
 $d_{mode} = 100$ km, and one EV
per two evacuees:

For Mariposa:
Population: 17k $\rightarrow$ #EVs: 8.5k
Charging energy: 8.5 MWh
Total charging time:
(8.5 MWh)/(2.5 MW) ~ 3 h

For Sacramento:
Population: 1.5M $\rightarrow$ #EVs: 750k
Charging energy: 750 MWh
Total charging time:
(750 MWh)/(36 MW) ~ 21 h

This result suggests that some of California's major cities, such as Sacramento, are underprepared for a large-scale EV evacuation with a short time window.

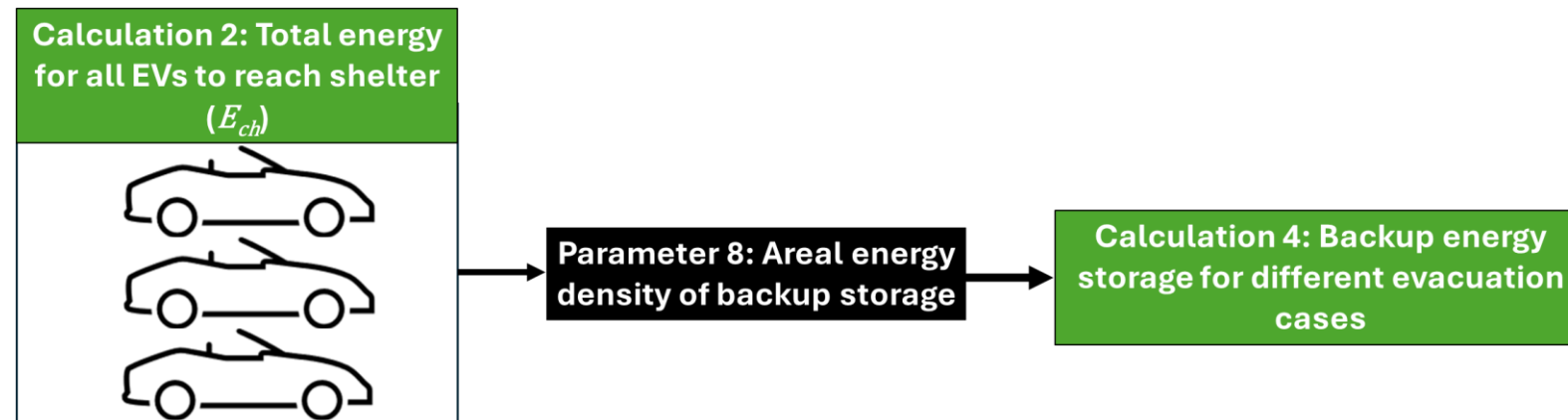


Part III: Backup lithium battery needs

How much land is required for backup lithium batteries to support EV evacuations during a grid failure?

Part III: Method to estimate backup storage size

We use the areal energy density of commercially available lithium battery storage systems to determine the size of the backup storage:



Based on available market products, the areal energy density for lithium batteries is 160-570 (kWh/m²)^{1,2}.

¹ TESLA Mega Pack

² LG Energy Storage Systems

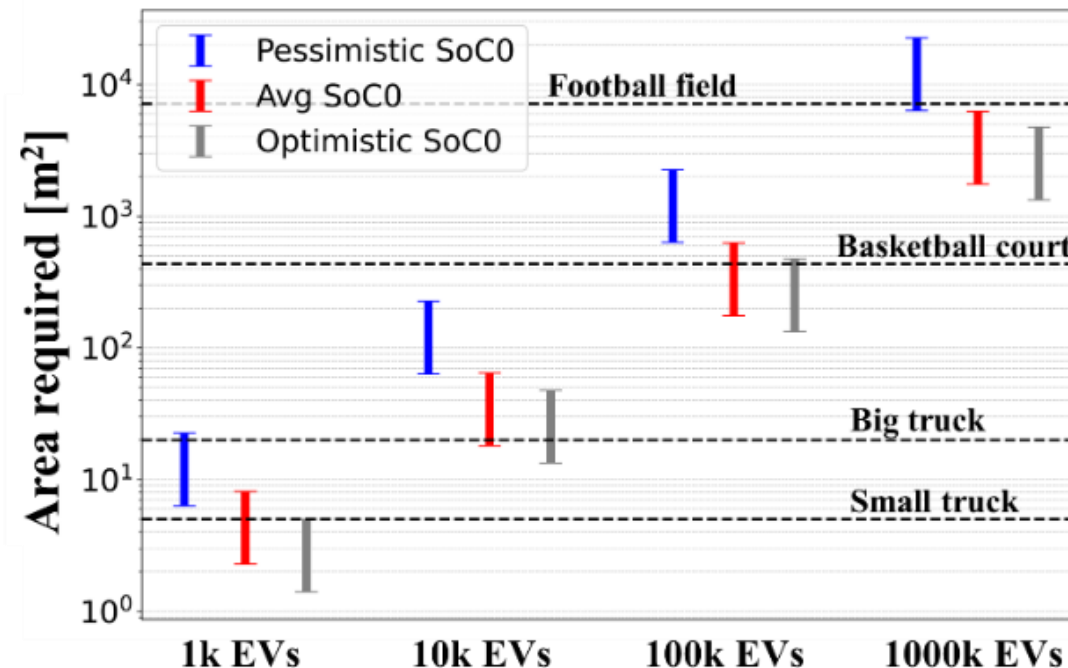


Part III: Backup storage size results

Size of backup lithium battery storage versus the scale of the evacuation:

$$d_{mode} = 100 \text{ km}$$

Assumption: No empty space between the storage products



Evacuating a small neighborhood with 1,000 EVs over a distance of 100 km could require a storage system small enough to be trucked in. However, larger evacuations, such as 1 million EVs over the same distance, may need storage the size of a football field.



Takeaways

- ✓ Small evacuations (1,000 EVs) over 100 km require MWh-level energy, while large evacuations (1M EVs) over the same distance need GWh-level energy.
- ✓ Many major California cities require additional chargers, particularly fast chargers, to handle evacuations within a short time window.
- ✓ In the event of grid failure, a truck-sized backup lithium battery system is sufficient for small evacuations (1,000 EVs), but a football-field-sized system is needed for larger evacuations (1M EVs) over 100 km.



ACKNOWLEDGMENT

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Thank you for your attention!

I will be happy to answer questions.